



SURYA PRATAP SARANGI

PROFILE

ECE undergraduate with experience in embedded systems, IoT development, and hardware-software integration. Skilled in Embedded C, Python, Raspberry Pi, Arduino, Linux, and communication protocols. Passionate about developing reliable embedded and connected solutions.

WORK EXPERIENCE

Operation Support System Engineer | JLK Technology

Apr 2026 – May 2026

- ✓ Diagnosed and resolved faults in electronic manufacturing systems.
- ✓ Performed hardware-level troubleshooting and system reliability analysis.
- ✓ Reduced downtime through structured fault isolation procedures.

IoT Trainer Associate | STEM Learning

Nov 2025 – Apr 2026

- ✓ Delivered training on microcontrollers, sensors, and IoT architecture.
- ✓ Guided students through hands-on embedded systems projects.

PROJECTS

Raspberry Pi-Based Real-Time Object Detection

Technologies: Python, Raspberry Pi, Pi Camera, OpenCV, Mobile Net

- ✓ Developed a real-time object detection system using Raspberry Pi and Pi Camera.
- ✓ Trained and deployed a Mobile Net-based model on the COCO dataset, achieving 89% detection accuracy.
- ✓ Integrated camera-based image acquisition and real-time inference for object tracking applications.
- ✓ Optimized the solution for edge computing environments with limited hardware resources.

Cactus Environmental Monitoring System

Technologies: Arduino, Embedded C, Temperature Sensor, Soil Moisture Sensor, Humidity Sensor, IoT

- ✓ Developed an environmental monitoring system to track key parameters affecting cactus growth, including temperature, humidity, and soil moisture.
- ✓ Interfaced multiple sensors with an Arduino-based embedded platform for real-time data acquisition.
- ✓ Implemented threshold-based monitoring to identify unfavourable environmental conditions.
- ✓ Improved plant health monitoring through continuous sensing and automated data collection.

Fire Detection System Using Arduino Nano BLE Sense

Technologies: Arduino Nano BLE Sense, Colour Sensor, Embedded C, Sensor Interfacing

- ✓ Designed an embedded system capable of detecting fire-related conditions and identifying object colours in real time.
- ✓ Integrated Arduino Nano BLE Sense with colour sensing hardware for environmental monitoring applications.
- ✓ Developed Embedded C firmware for sensor data acquisition, processing, and event detection.
- ✓ Implemented alert generation based on sensor readings to enhance safety and monitoring capabilities.

CONTACT

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SKILL

Programming Language:

C, Embedded C, C++, Python

Communication Protocol:

UART, SPI, I2C, SERIAL
COMMUNICATION, MQTT

OPERATING SYSTEMS:

LINUX, UBUNTU, RASPBIAN,
WINDOWS

Tools & Platforms:

Arduino IDE, Proteus, Git, VS
Code

IoT & Edge Computing:

IoT System Design, Real-Time
Data Acquisition, Edge AI
Applications, Wireless Sensor
Integration

Libraries & Frameworks:

OpenCV, NumPy

Core Concepts:

Microcontrollers, Analog &
Digital Electronics, Signal
Conditioning, Interrupts,
ADC/DAC, Embedded
Debugging, PCB Basics,
Electronics Basic, Sensors

Used Microcontroller:

Arduino, ESP32/ESP8266,
STM32, RASBERRY PI

EDUCATION

**Centurion University of
Technology and Management,
Bhubaneswer**

B.Tech – Electronics &
Communication Engineering
2022 – 2026
CGPA: 8/10